Code: 20A33701a

R20

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

MACHINE LEARNING FOR UNSTRUCTURED DATA

(CSE (AI&ML))

Time: 3 hours Max. Marks: 70

PART – A

(Compulsory Question)

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1 Answer the following: (10 X 02 = 20 Marks)

(a)	Define Text Mining.	2M
(b)	What is the importance of background knowledge in Text Mining?	2M
(c)	List the applications of Text Categorization.	2M
(d)	Outline the metrics used to evaluate the quality of clustering results.	2M
(e)	Define Anaphora Resolution.	2M
(f)	Describe the basic principle behind Maximal Entropy Modeling.	2M
(g)	Mention the different types of accessing constraints.	2M
(h)	What architectural considerations are important when designing visualization tools for text mining?	2M
(i)	How does path differ from cycle in graphs?	2M
(j)	List the software packages used for Link Analysis.	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2		Explain the core Text Mining operations and their importance in analyzing unstructured text.	10M
		OR	
3	(a)	Discuss how task-oriented approaches influence the Text Mining process.	5M
	(b)	What are text mining query languages, and how do they help in extracting meaningful information from unstructured data.	5M
4	(a)	Distinguish between Classification and Clustering in Text Mining.	5M
	(b)	Discuss two clustering algorithms used in Text Mining. How can they be applied to analyze textual data?	5M
		OR	
5	(a)	Describe the document representation methods used in Text Categorization.	5M
	(b)	Summarize the Clustering tasks in Text Analysis.	5M
6	(a)	Illustrate the architecture of Information Extraction systems.	5M
	(b)	Analyses the applications of HMM in Textual Analysis.	5M
		OR	
7	(a)	What are inductive algorithms and how do they differ from rule based algorithms in information extraction?	5M
	(b)	Analyses the applications of CRFs in Textual Analysis.	5M
8	(a)	How Simple Specification Filters are used at the presentation layer? Give examples.	5M
	(b)	Discuss common visualization approaches that can be applied in Text Mining to reveal patterns in data.	5M
		OR	
9	(a)	How does the presentation layer interact with the underlying query language in an Information Extraction system?	5M
	(b)	Outline the visualization techniques in Link Analysis.	5M
10		Distinguish different layout techniques used in Link Analysis with their merits and demerits.	10M
		OR	
11		Describe how text mining is used in life sciences research, particularly in mining biological pathway information using systems like Gene Ways.	10M

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